Adaptive Pathways — data report

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Repository: https://github.com/Muhammadabrarrayhan2/adaptive-pathways-appinventor

Generated by `python run_pipeline.py report`. All numbers below are recomputed from `data/` on every run.

1. Corpus

source	total	in_analysis
github	374	371
official	14	14
gallery	12	11

- Projects stored: **400**; used in analysis: **396** (excluded: 3 structural near-duplicates, 1 empty, 0 parse errors).
- Downloads rejected before storage: `{'not_zip': 2, 'too_large': 1}` (`not_app_inventor` = other software's `.aia` files, e.g. Adobe Illustrator actions).
- Licences (top): `{'MIT': 253, 'GPL-3.0': 66, 'Apache-2.0': 21, 'CC0-1.0': 18, 'CC BY-SA 4.0 (MIT App Inventor)': 14, 'CC BY-SA 4.0 (App Inventor Gallery terms)': 12, 'Unlicense': 4, 'GPL-2.0': 4, 'BSD-2-Clause': 2, 'CC-BY-4.0': 1}`

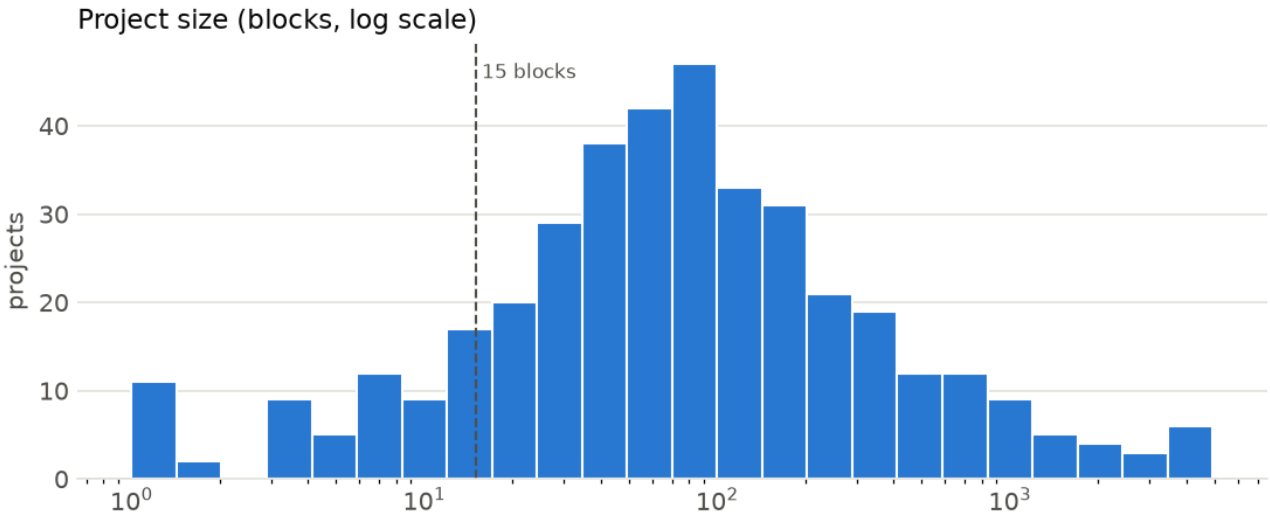
2. Parser quality

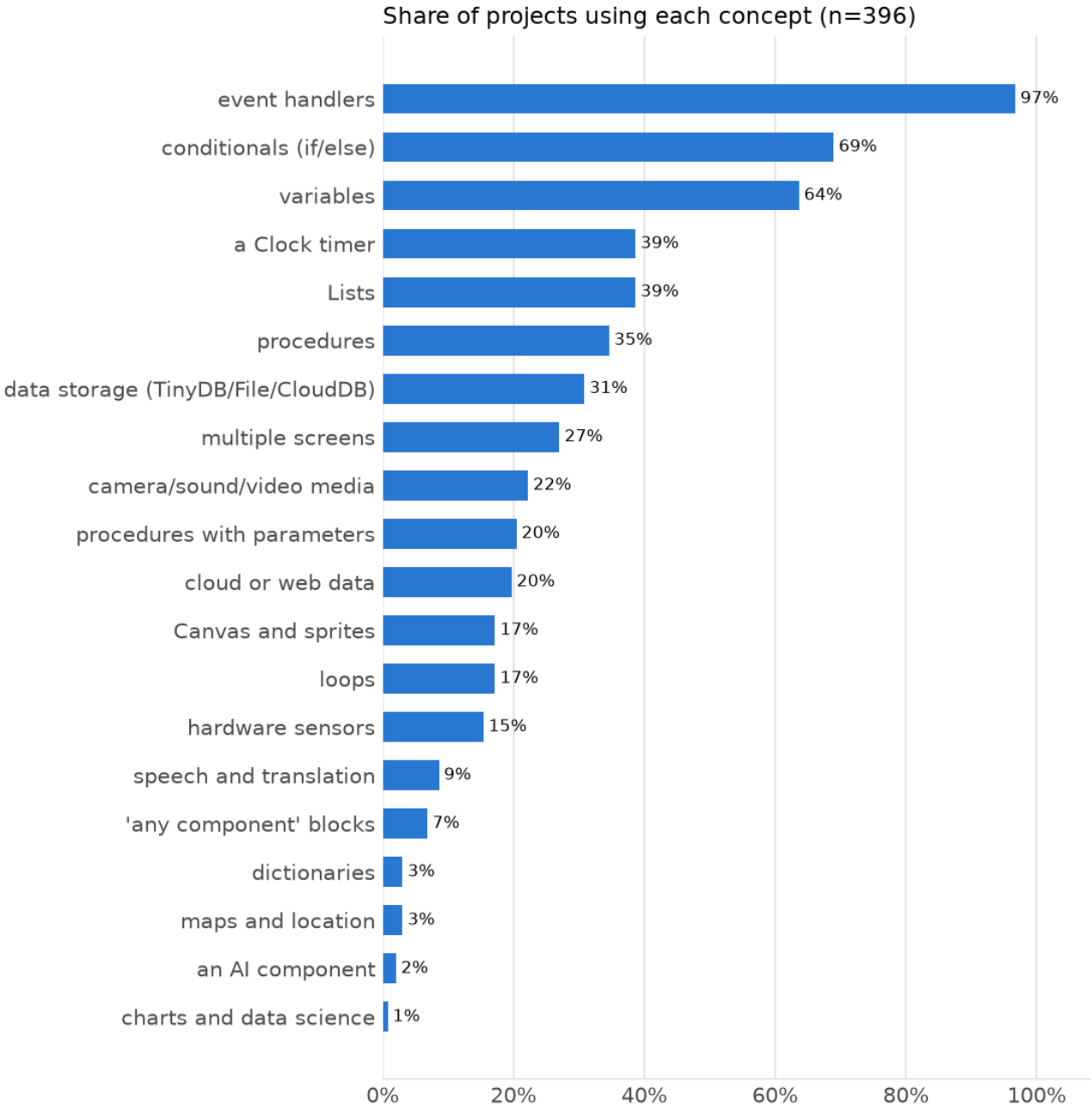
- aiatools parsed **98.5%** of projects; the rest used the direct XML/JSON reader.
- Agreement between aiatools and the direct reader: blocks **99.7%**, components **99.2%**.
- 6 projects use the pre-2013 block format (`Button1_Click`); they are normalised to the modern vocabulary.
- Known aiatools issue handled: project-level `AIAFile.blocks()` returns only one screen's blocks in multi-screen projects; the pipeline aggregates `screen.blocks()` per screen instead.

3. Comparison with the published Gallery baseline

metric	this corpus	Gresse von Wangenheim et al.
mean_blocks	261.551	162.500
pct_under_15_blocks	0.144	0.250
pct_with_events	0.967	0.900
pct_with_loops	0.172	0.060
pct_with_button_or_label	0.960	0.850

Differences are expected: this corpus over-samples curated sources (official tutorials, featured Gallery apps, GitHub repositories), so it skews larger and more structured than the full Gallery.


Project size (blocks, log scale)
15 blocks
projects
0
10
20
30
40
10^0
10^1
10^2
10^3


Share of projects using each concept (n=396)
event handlers 97%
conditionals (if/else) 69%
variables 64%
a Clock timer 39%
Lists 39%
procedures 35%
data storage (TinyDB/File/CloudDB) 31%
multiple screens 27%
camera/sound/video media 22%
procedures with parameters 20%
cloud or web data 20%
Canvas and sprites 17%
loops 17%
hardware sensors 15%
speech and translation 9%
'any component' blocks 7%
dictionaries 3%
maps and location 3%
an AI component 2%
charts and data science 1%
0%
20%
40%
60%
80%
100%

4. Clustering (RQ1)

Primary solution: **kproto, k = 3** (silhouette 0.133, stability ARI 0.989) — rule: max silhouette over (method, k) with stability ARI >= 0.6 and min cluster size >= 3.

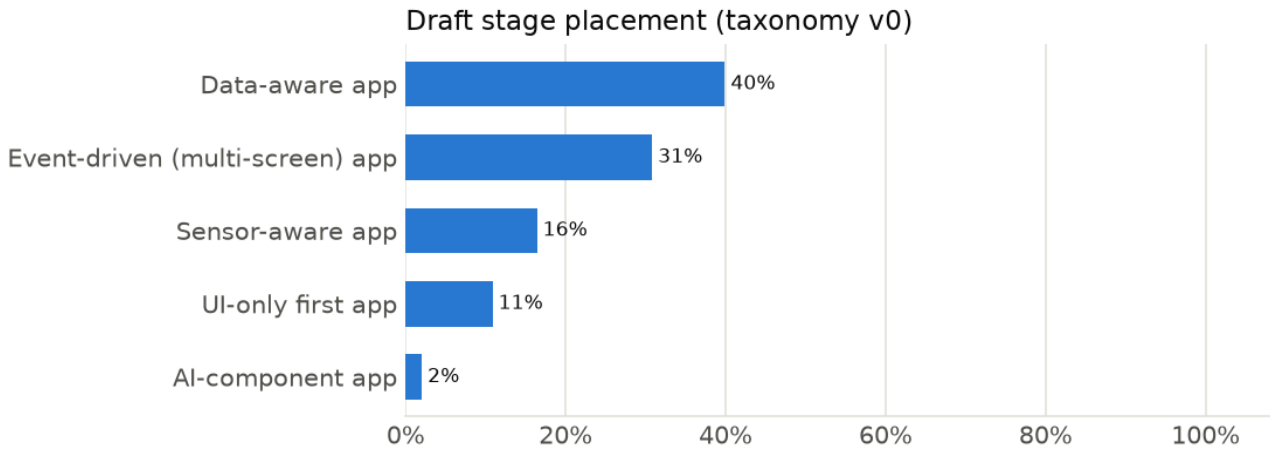
Silhouette values around 0.1 indicate overlapping rather than sharply separated groups: learner projects form a continuum, which is consistent with treating stages as a progression rather than discrete types.

k	silhouette_hier	silhouette_kproto	kproto_seed_stability_ari	hier_bootstrap_stability_ari	hier_vs_kproto_ari	min_cluster_size_hier	min_cluster_size_kproto
3	0.095	0.133	0.989	0.457	0.358	100	81
4	0.107	0.132	0.690	0.370	0.402	59	40
5	0.102	0.114	0.914	0.528	0.362	13	24
6	0.108	0.103	0.647	0.488	0.339	3	25
7	0.119	0.100	0.569	0.497	0.325	3	11
8	0.127	0.072	0.572	0.565	0.268	3	18
9	0.129	0.104	0.558	0.557	0.278	3	11

5. Draft stages vs. data-driven clusters

ARI between the primary clusters and rule-based draft stages: **0.125** (1 = identical partitions, 0 = chance). Low purity clusters are where the teacher review should focus.

cluster	n	majority_stage	majority_stage_name	purity	stage_mix
0	104	1	UI-only first app	0.413	{"ui_first_app": 43, "event_driven": 35, "data_aware": 14, "sensor_aware": 11, "ai_component": 1}
1	81	3	Data-aware app	0.741	{"data_aware": 60, "sensor_aware": 14, "event_driven": 4, "ai_component": 3}
2	211	3	Data-aware app	0.398	{"data_aware": 84, "event_driven": 83, "sensor_aware": 40, "ai_component": 4}



6. Next-step catalog

56 resources (13 with concepts measured from their own solution `.aia`).

7. Limitations

- Gallery listings are only partly reachable without login (featured apps); Gallery projects are collected only through public listings. GitHub repositories (open licences only) over-represent classroom and tutorial work.
- Stage labels are a rule-based draft (taxonomy v0) awaiting the two-teacher review in `taxonomy/stage_labels.yaml`.
- No telemetry, no learner outcomes: everything here is structural and descriptive.